



Lingwistyka matematyczna

Zadania:

1.

Quenya (Roxana Dincă, RoLO 2013)

Here are some Quenya numbers:

<i>neldë</i>	3	<i>enquë yucainen</i>	26
<i>canta</i>	4	<i>minë nelcainen</i>	31
<i>lempë</i>	5	<i>cancainen</i>	40
<i>otso</i>	7	<i>atta tolcainen</i>	82
<i>tolto</i>	8	<i>atta tolcainen tuxa</i>	182
<i>nelcëa</i>	13	<i>nertë nelcainen lemtuxa</i>	539
<i>encëa</i>	16		

Problem 9.1a Write in numerals:

- | | |
|-----------------------------|---------------------------|
| a. <i>tolcëa</i> | d. <i>lempë tolcainen</i> |
| b. <i>enquë cancainen</i> | e. <i>tuxa</i> |
| c. <i>cancainen neltuxa</i> | |

Problem 9.1b Write in Quenya: 1, 70, 192, 385.

2.

Embera Chami (Vlad A. Neacșu, PLO 2022)

Here are two equalities in Embera Chami:

- (1) $umbea + huasoma kwimane = omme huasoma omme$
(2) $omme huasoma kwimane + huasoma abba = kwimane huasoma$

Problem 9.2a Write the equalities above with numerals.

Problem 9.2b Write in Embera Chami: 1, 5, 17, 23.



3.

Yup'ik (Kai Low Rui Hao, UKLO 2017)

Yup'ik people have an interesting concept when it comes to counting – the words for the numbers can be broken down into meaningful parts which may be related to body parts. For example, the word for 5, *talliman*, means 'arm' and the word for 6, *arvinlegen*, means 'cross over', as you need to change hands to go on counting.

The Yup'ik people often include geometry in the border patterns of their traditional garments, called "parkas". One such pattern comes from a 3×3 square, as represented below. This is a magic square, constructed by placing the digits 1 to 9 within the cells such that the sum of all the digits in every row, column, and diagonal is the same.

	1	2	3
a		9	
b			
c			

To help you fill in the magic square, the following clues are given in Yup'ik.
*Hint: 294 in Yup'ik is *yuinaat qula cetaman qula cetaman*.*

- Rows:

- Yuinaat yuinaq cetaman qula malruk*
- Yuinaat akimiaq malruk akimiaq malruk*
- Yuinaat yuinaak malruk akimiaq atauciq*

- Columns:

- Yuinaat yuinaq atauciq akimiaq pingayun*
- Yuinaat yuinaak malruk yuinaat malrunlegen qula atauciq*
- Yuinaat qula pingayun akimiaq atauciq*

Problem 9.3a Fill in the numbers missing from the magic square above. One digit is already given ($a_2 = 9$).

Problem 9.3b Write in Yup'ik the number given in the diagonal from top left (the number formed by the digits a_1 - b_2 - c_3).



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4.

Umbu-Ungu (Ksenia Gilyarova, IOL 2012)

Umbu-Ungu	Umbu-Ungu
10 <i>rureponga talu</i>	35 <i>tokapu rureponga yepoko</i>
15 <i>malapunga yepoko</i>	40 <i>tokapu malapu</i>
20 <i>supu</i>	48 <i>tokapu talu</i>
21 <i>tokapunga telu</i>	50 <i>tokapu alapunga talu</i>
27 <i>alapunga yepoko</i>	69 <i>tokapu talu tokapunga telu</i>
30 <i>polangipunga talu</i>	79 <i>tokapu talu polangipunga yepoko</i>
	97 <i>tokapu yepoko alapunga telu</i>

Problem 9.4a Write in numerals:

- tokapu polanigpu*
- tokapu talu rureponga telu*
- tokapu yepoko malapunga talu*
- tokapu yepoko polangipunga telu*

Problem 9.4b Write in Umbu-Ungu: 13, 66, 72, 76, 95.

5.

Problem 9.6

Yoruba (Harold Somers, UKLO 2020)

Here are some numbers in Yoruba:

<i>èji</i>	2	<i>ẹ́ẹ̀rìndilogóji</i>	36
<i>ẹ̀rin</i>	4	<i>ẹ̀rìndogóji</i>	44
<i>àrun</i>	5	<i>àádorin</i>	70
<i>ẹ̀rinlá</i>	14	<i>ẹ́ẹ̀tádilogórin</i>	77
<i>eéjìdilogun</i>	18	<i>ẹ̀tádogórin</i>	83

Problem 9.6a Write in numerals:

- àádota*
- àrùndogórin*
- aárùndilogórin*
- ẹ̀tádogórun*
- òkándilogóji*

Problem 9.6b Write in Yoruba: 12, 45, 57, 90, 99.



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6.

Czech (Mirjam Fried, Princeton)

Here are some examples of how to tell the time in Czech:

<i>za pět minut osm</i>	'five minutes to eight'
<i>za deset minut osm</i>	'ten minutes to eight'
<i>čtvrt na osm</i>	'quarter past seven'
<i>za sedm minut osm</i>	'seven minutes to eight'
<i>za osm minut čtvrt na osm</i>	'seven minutes past seven'
<i>za deset minut čtvrt na sedm</i>	'five minutes past six'
<i>půl osmé</i>	'half past seven'
<i>půl deváté</i>	'half past eight'
<i>za deset minut půl šesté</i>	'twenty minutes to five'
<i>čtvrt na deset</i>	'quarter past nine'

Problem 9.7a Translate into Czech:

1. 'twenty-three minutes past five'
2. 'ten minutes to nine'

7.

Swahili (Nilai Sarda, PLO 2014)

Here are some Swahili phrases and their English translations *in random order*:

- | | |
|--|-------------------------|
| 1. <i>jumamosi, saa moja usiku</i> | A. 'Sunday, 1.00 AM' |
| 2. <i>jumapili, saa tatu na robu asubuhi</i> | B. 'Sunday, 7.30 AM' |
| 3. <i>jumamosi, saa saba usiku</i> | C. 'Sunday, 9.15 AM' |
| 4. <i>jumamosi, saa mbili na robu usiku</i> | D. 'Tuesday, 12.15 PM' |
| 5. <i>jumanne, saa tano na nusu usiku</i> | E. 'Tuesday, 11.30 PM' |
| 6. <i>jumanne, saa sita na robu asubuhi</i> | F. 'Saturday, 10.30 AM' |
| 7. <i>jumamosi, saa nne na nusu asubuhi</i> | G. 'Saturday, 7.00 PM' |
| 8. <i>jumapili, saa moja na nusu asubuhi</i> | H. 'Saturday, 8.15 PM' |

Problem 9.8a Determine the correct correspondences.

Problem 9.8b Translate into English:

9. *jumatano, saa moja na robu asubuhi*
10. *jumapili, saa nne na nusu asubuhi*



8.

Danish (Michael Swan, UKLO 2012)

Here are some numbers written in Danish:

<i>tre</i>	3	<i>tredive</i>	30
<i>fire</i>	4	<i>fyrre</i>	40
<i>fem</i>	5	<i>syvoghalvtreds</i>	57
<i>seks</i>	6	<i>tres</i>	60
<i>syv</i>	7	<i>otteoghalvfjerds</i>	78
<i>tyve</i>	20	<i>firs</i>	80

Problem 9.9a Write in numerals:

- | | |
|---------------------------|---------------------------|
| a. <i>treogtyve</i> | d. <i>femoghalvfjerds</i> |
| b. <i>seksoghalvtreds</i> | e. <i>syvoghalvfems</i> |
| c. <i>fireogtres</i> | |

Problem 9.9b Write in Danish: 8, 27, 36, 65, 98.

9.

Waorani (Dragomir R. Radev, UKLO 2012)

Here are some equalities written in Waorani. Each sequence printed in bold represents one number from 1 to 10.

- (1) $\text{mēña mēña mēña mēña} + \text{mēña go mēña} = \text{āēmāēmpoke go aroke} \times 2$
- (2) $\text{aroke}^2 + \text{mēña}^2 = \text{āēmāēmpoke}$
- (3) $\text{āēmāēmpoke go aroke}^2 = \text{mēña go mēña} \times \text{āēmāēmpoke mēña go mēña}$
- (4) $\text{mēña} \times \text{āēmāēmpoke} = \text{tipāēmpoke}$

Problem 9.11a Write in Waorani the numbers from 4 to 10.